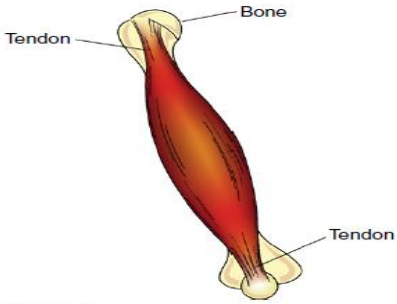
Equilibrium and Stability

- **2. Skeletal Muscles:**



Drawing of a muscle.

- Skeletal muscles produce skeletal movements. They consist of thousands of parallel fibers wrapped

in a flexible sheath that narrows into tendons at both ends. These tendons, made of strong tissue,

attach the muscle to the bones.

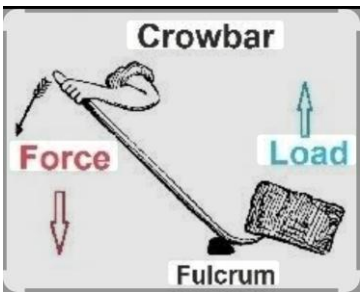
- Most muscles taper to a single tendon, but some end in two tendons (biceps) or three tendons (triceps).
- Each end of a muscle attaches to a different bone, allowing movement at the joints.
- Leonardo da Vinci noted: "The muscles always begin and end in the bones that touch

one another… It is the function of the muscles to pull and not to push except in the cases of the genital member and the tongue."

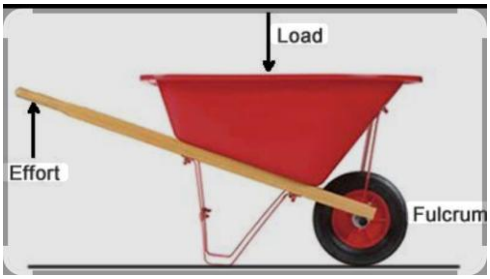
- Contraction Mechanism:
  When muscle fibers receive an electrical stimulus from nerve endings, they contract, shortening the muscle and pulling on the two attached bones.
- Muscle Force:
  The maximum force a muscle can exert is proportional to its cross-sectional area. It is estimated as: $7 * 10^6 dy\, n/cm^2 = 7 * 10^5 Pa\ 102\ \ Ib/in^2$
- **3. Levers:**
- A lever is a rigid bar free to rotate about a fixed point called the fulcrum.
- Levers are used to lift loads advantageously and transfer motion.

- Three Classes of Levers:

  1. Class 1: Fulcrum between applied force and load.
     Example: Crowbar.

- 

- 2. Class 2: Fulcrum at one end, force at the other, load in between.
  Example: Wheelbarrow.

- 

  3. Class 3: Fulcrum at one end, load at the other, force applied in between.
     Most limb movements in animals are Class 3 levers.

- Equilibrium Equations:
  The force (F) required to balance a load of weight W is:

$$F = \frac{W.d_1}{d_2}$$

  d_1 : distance from fulcrum to line of action of F.
  d_2 : distance from fulcrum to line of action of W.

- Mechanical Advantage (M):

$$M = \frac{W}{F} = \frac{d_2}{d_1}$$

- Relationship between displacements (L) and velocities (v):

$$\frac{l_1}{l_2} = \frac{d_1}{d_2}\ ,\ \frac{v_1}{v_2} = \frac{d_1}{d_2}$$

$l_1 . v_1$: displacement and velocity of the load.
$l_2 . v_2$: displacement and velocity of the point where force is applied

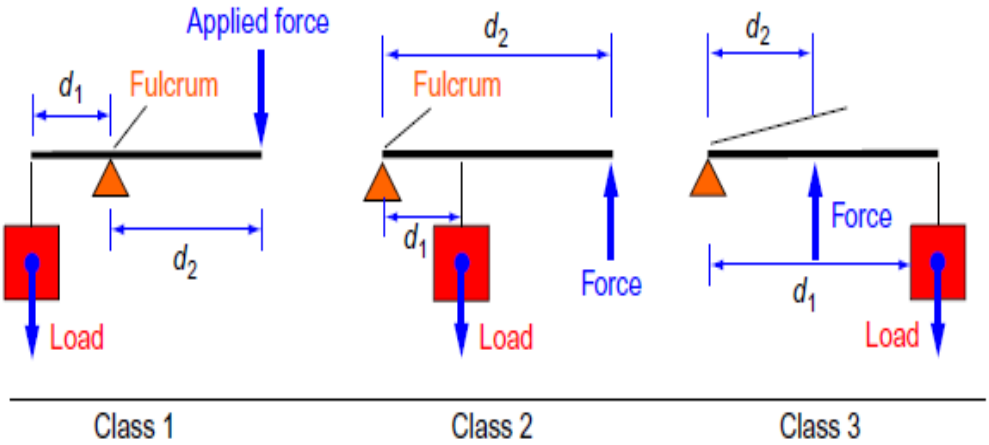
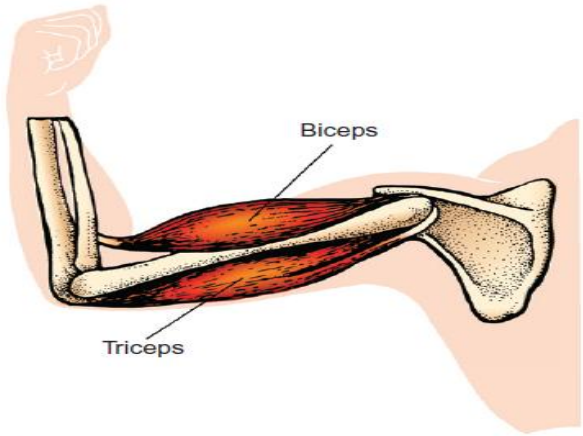


FIGURE 1.9 The three classes of lever.

- **Application: The Elbow Joint:**
- The two main muscles controlling elbow movement are:
  1. Biceps: Contraction closes (flexes) the elbow.
  2. Triceps: Contraction opens (extends) the elbow.
- **Analysis**: When holding a weight (W) in the hand with the elbow bent at 100^0(Fig. 12), the forearm acts as a Class 3 lever:
  · Fulcrum: elbow joint.
  · Load: weight W in the hand.
  · Force: pulling force of the biceps F_B applied to the forearm.



The elbow.

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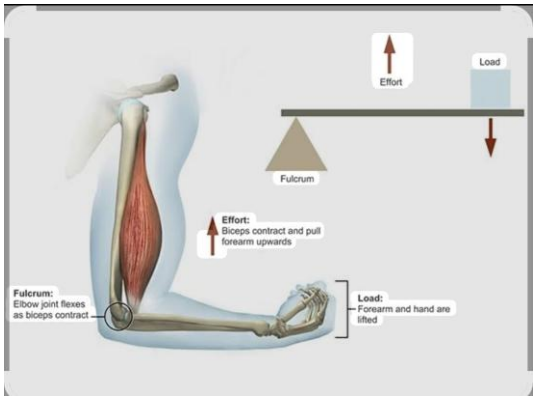
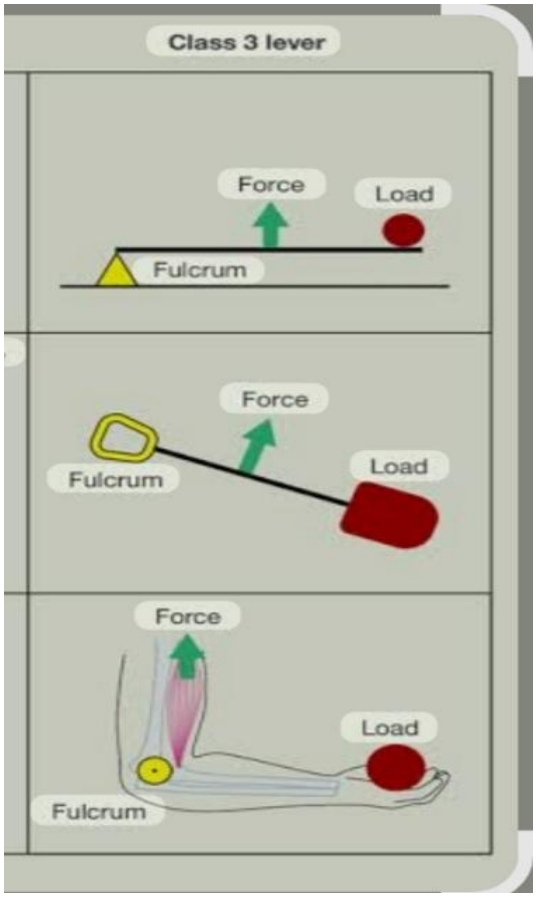




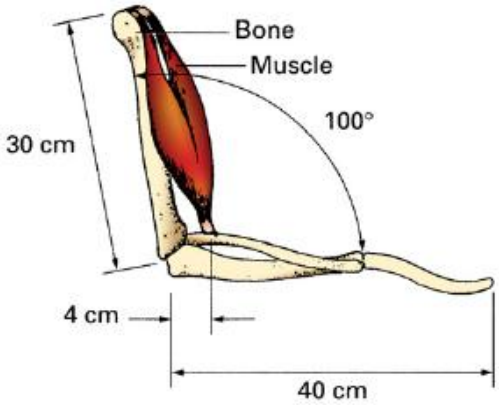
- We note that:
  d_1: (distance from fulcrum to biceps force) is larger than d_2 (distance from fulcrum to load). Therefore, mechanical advantage is less than 1 (M<1). This means the biceps force F_{B} must be greater than the load W. However, this allows a greater range of motion and speed of the hand for a small muscle contraction.

- The elbow joint is an example of a Class 3 lever. When holding a weight in the hand, the biceps muscle exerts an upward pull on the forearm to balance the weight.

- Conclusion: The elbow experiences very large forces; the muscle force is ten times the load, allowing high speed of hand motion at the expense of muscular effort.

- Summary
  - Skeletal muscles pull on bones via tendons, causing movement.
  - Joints can be analyzed using the lever model.
  - The elbow represents a third-class lever, where the muscular force required is greater than the load, but it provides a greater range and speed of motion.
  - The fundamental equations relate force, load, distances, displacements, and velocities, and determine the mechanical advantage of the system.

- **Example**



(B)

A simplified drawing of (A).

- **Application: The Hip joint:**

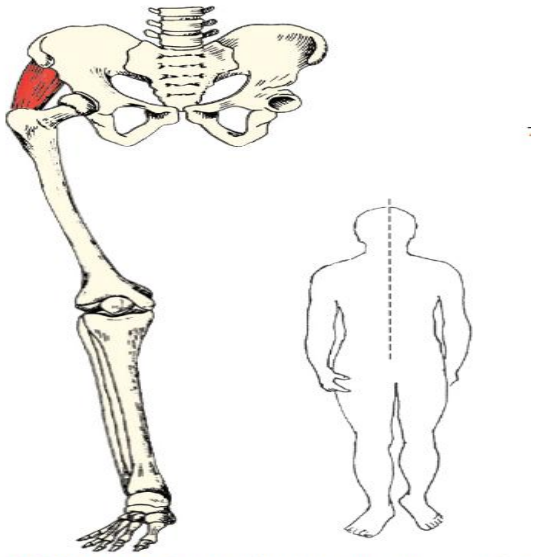
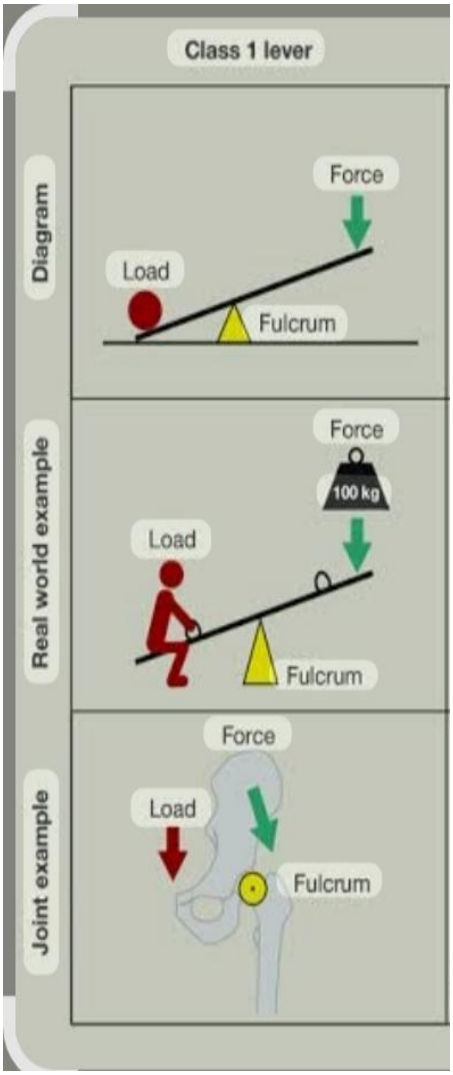


FIGURE 1.14 (A) The hip. (B) Its lever representation



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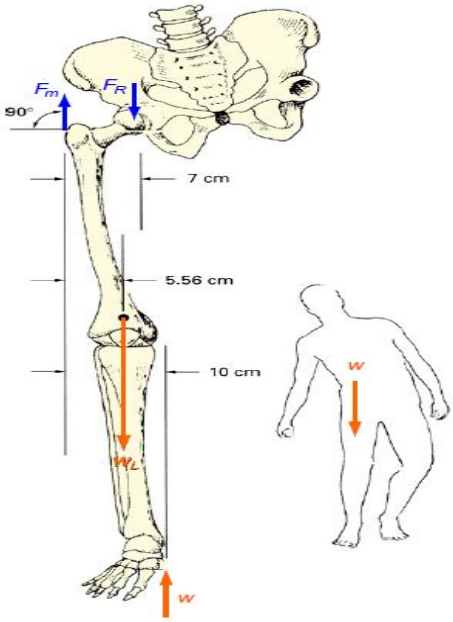
- The hip joint and its simplified lever representation, can be represented as a Class 1 lever

(fulcrum between muscle force and load). When standing on one foot (as in slow walking), large forces act on the joint.

- The hip is stabilized in its socket by a group of muscles, which is represented in (Fig) as

a single resultant force (F_M). When a person stands erect, the angle of this force is about 71° with respect to the horizon. (W_L) represents the combined weight of the leg, foot, and thigh. Typically, this weight is a fraction (0.185) of the total body weight W (i.e $(W_L = 0{:}185W)$. The weight $(W_L$ is assumed to act vertically downward at the midpoint of the limb

- **Application: The Hip joint:**
- **Limping**:



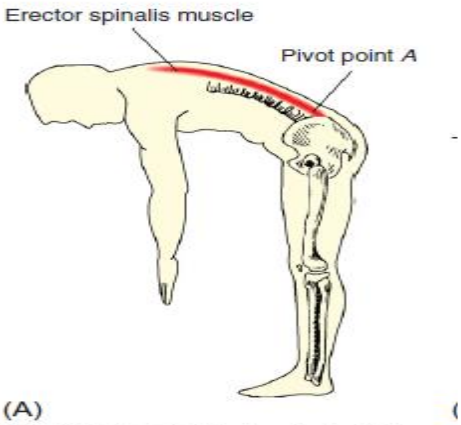
Walking on an injured hip.

- With an injured hip, the person leans toward the injured side when stepping on that foot.
- This shifts the body's center of gravity almost directly above the joint, reducing the torque

due to weight. Calculations show the muscle force becomes $(F_M = 0.47\ W)$, and the hip joint

force becomes 1.28W (instead of 2.37W), a significant reduction.
This is a significant reduction from the forces applied during a normal one legged stance.

- **Application: The Back:**



❖ FIGURE 1.16 (A) The bent back. (B) Lever

❖ When the trunk bends forward, the spine pivots mainly on the fifth lumbar vertebra (fulcrum A).

The back is represented as a lever (Fig).

❖ **Data:**

❖ Trunk weight (W_1) uniformly distributed, represented by a weight at the middle of the lever arm.

❖ Head and arms weight (W_2) at the end of the lever arm.

❖ Erector Spinae muscle holds the back, attached at a point two-thirds up the spine. The angle between spine and muscle is about 12°.

❖ For a 70-kg person: ($W_1 = 320$ N (72Ib), $W_2 = 160N$ (36 Ib)

❖ This shows that very large forces act on the lower back, explaining why backaches often originate there. The forward-bending posture to lift a weight is not recommended as it increases the load on the vertebrae.

Static Forces in the human body (Conts.)

❖ **3. Levers:**

❖ **Application: Standing Tip-toe on One Foot:**

❖ IGURE 1.17 (A) Standing on tip-toe. (B) Lev

- When standing on tip-toe on one foot (Fig), the body is supported by the ground reaction force at point A. This is a Class 1 lever, with the fulcrum at the contact point of the tibia. The balancing force is provided by the calf muscle (soleus/gastrocnemius) connected to the heel via the Achilles tendon.
- Calculations (mentioned in text):
  - Compressive force on the tibia: 3.5 × body weight (W).
  - Tension force on the Achilles tendon: 2.5 × W.
  - Example: For a person weighing 686 N, the compressive force on the tibia is about 2401 N, and the tendon tension is about 1715 N. This position is quite strenuous.

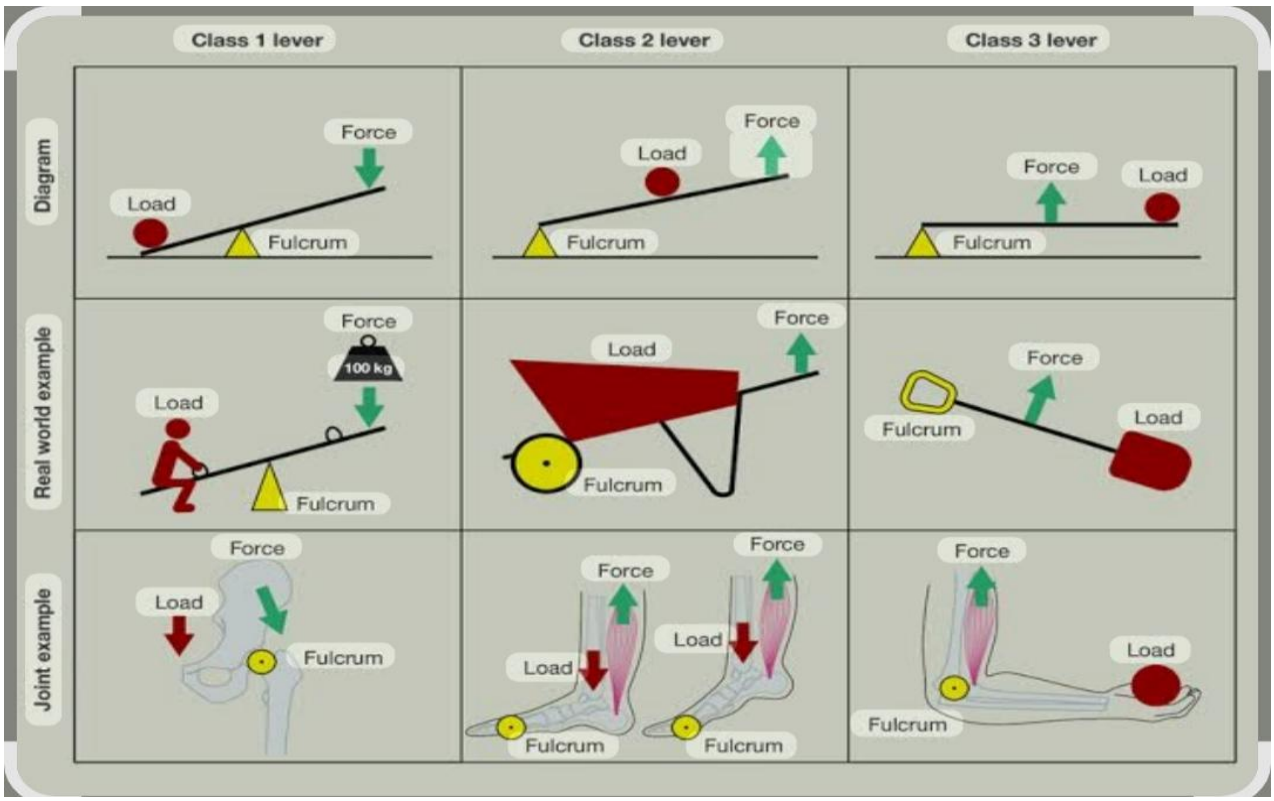
Dynamic Aspects of Posture

- The human body is not completely static; it is a dynamic system continually responding to internal and external stimuli.
- **Maintaining Balance While Standing:**
  - The body's center of gravity is at about half the height above the soles, so even a slight displacement can cause toppling.
  - Experiments show that standing erect requires continuous swaying (back-forth and left-right) to keep the center of gravity over the base of support.
  - Small perturbations (<1.5 cm) are compensated by ankle movements; larger ones require hip movements.
- **Walking:**
  Maintaining balance during walking is more complex as support shifts from one foot to another.The nervous system performs a highly complex task to keep upright, especially when tripping or slipping.

Dynamic Aspects of Posture

- **Information Sources for Balance:**
  1. Vision.
  2. Vestibular system in the inner ear, monitoring head movement and position.
  3. Somatosensory system, monitoring position and orientation of body parts.
- With age, the efficiency of these systems decreases, increasing the risk of falls and injuries.
- **Interconnectedness of the Musculoskeletal System:**

All muscles and bones are interconnected; any change in muscle tension or limb position in one part must be compensated elsewhere. For example, excessive tightness of the large muscles at the front of the legs may pull the torso forward, requiring compensatory tightening of back muscles, potentially causing pain in the lower back.

Conculsion

- Muscles and skeleton act as lever systems, mostly Class 3 (e.g., elbow), trading force for speed and range of motion.
  Skeletal muscles pull on bones via tendons, causing movement.
- Forces within joints are very large, often multiples of body weight
- Static analysis gives estimates of these forces, but dynamics are essential for understanding real balance and movement
- Human body design shows trade-offs between strength, speed, and precision, with great complexity in neuromuscular control.